

Nonlocal Wasserstein Entropy Flows, Fractional Generators, and PL/KL Inequalities

Gabriel Peyré

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Abstract

This note asks when the flow of $\alpha \mapsto \text{KL}(\alpha \mid \beta)$ generated by a reversible jump kernel satisfies a Polyak–Łojasiewicz (PL), or more generally a Kurdyka–Łojasiewicz (KL), inequality. At the semigroup level, the exact condition is the *modified logarithmic Sobolev inequality* (MLSI). It compares relative entropy with the nonlocal Fisher information $\mathcal{E}_K(r, \log r)$, where $r = d\alpha/d\beta$. It becomes a genuine metric PL inequality once the logarithmic-mean transport metric has this Fisher information as the squared entropy slope. This distinction is automatic in standard finite-state and translation-invariant frameworks, but requires care for singular target-dependent kernels. In every case the answer is not a property of β alone: it depends essentially on the jump kernel.

Several useful sufficient conditions are established: positive geodesic curvature of the entropy, a Gross or weak log-Sobolev inequality for the jump Dirichlet form, a Doeblin minorization, and a spectral gap on classes of uniformly bounded densities. An exact two-level test quantifies bottleneck obstructions. A particularly clean positive construction is obtained by subordinating the reversible Langevin semigroup. If β satisfies the ordinary log-Sobolev inequality with constant λ and ϕ is a Bernstein function, then the nonlocal generator $-\phi(-L_\beta)$ satisfies MLSI with constant at least $\phi(2\lambda)/2$. For the standard Gaussian and $\phi(q) = q^{s/2}$, this gives a Gaussian-reversible fractional Ornstein–Uhlenbeck flow with MLSI constant at least $2^{s/2-1}$. This is not the same as adding a translation-invariant fractional Laplacian to an Ornstein–Uhlenbeck drift, whose invariant law is stable rather than Gaussian when $s < 2$ and which is generally not reversible. Direct stable-like reversible forms provide a second positive construction for the stretched-exponential family $e^{-(1+|x|^2)^q}$, including log-concave targets with tails heavier than Gaussian that fail the ordinary local Gross LSI.

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1 Question and main conclusions

Let β be a probability measure and consider

$$f(\alpha) = \text{KL}(\alpha \mid \beta). \quad (1)$$

For the ordinary 2-Wasserstein geometry, the squared metric slope of (1) is the relative Fisher information. The classical logarithmic Sobolev inequality is therefore exactly the Wasserstein–PL inequality for f . The same dictionary survives for the nonlocal Wasserstein geometry generated by a reversible jump kernel, but the Fisher information becomes a pair-space quantity.

The conclusions of this note are as follows.

1. For a fixed reversible kernel K , the exact semigroup condition is the modified logarithmic Sobolev inequality (MLSI)

$$\mathcal{I}_K(r) \geq 2\kappa \text{Ent}_\beta(r), \quad r = \frac{d\alpha}{d\beta}. \quad (2)$$

It is equivalent to uniform exponential entropy decay of the jump semigroup. It is also the metric PL inequality whenever the nonlocal transport metric satisfies the entropy-slope identity.

2. No assumption on β alone can imply (2). The kernel can be slowed down, disconnected, or made arbitrarily weak in the tails without changing β .
3. Mere log-concavity of β does not settle the question. Strong log-concavity does give a positive result for a canonical family of nonlocal kernels: subordinate the reversible Langevin generator.

4. The translation-invariant fractional Laplacian on $\mathbb{R}^d$ is reversible with respect to Lebesgue measure, not a confining probability measure. It therefore does not directly define the gradient flow of $\text{KL}(\cdot \mid \beta)$ for a Gaussian or another nonconstant target.
5. For $\beta = \mathcal{N}(0, I_d)$, the fractional power of the Ornstein–Uhlenbeck generator is a genuine Gaussian-reversible nonlocal model. Its jump kernel is explicit as a mixture of Mehler kernels, and it satisfies a quantitative MLSI.
6. A direct stable-like reversible kernel also satisfies MLSI for $\beta_q \propto e^{-(1+|x|^2)^q}$ for every $q > 0$. The range $1/2 \leq q < 1$ is especially informative: these targets are log-concave but fail the ordinary local Gross LSI, while the long-jump form remains entropy-coercive.

The word “KL” is ambiguous in this discussion. We write KL for the Kullback–Leibler divergence and reserve “power-KL inequality” for a Kurdyka–Łojasiewicz slope inequality.

2 Reversible jump geometry and entropy dissipation

2.1 Kernel, edge measure, and Dirichlet form

Let $(\mathcal{X}, \beta)$ be a probability space. A jump kernel $K(x, dy)$ is *reversible* with respect to β when the edge measure

$$J_\beta(dx, dy) := \beta(dx)K(x, dy) \quad (3)$$

is symmetric under $(x, y) \mapsto (y, x)$. The generator and its Dirichlet form are

$$L_K h(x) := \int_{\mathcal{X}} (h(y) - h(x)) K(x, dy), \quad (4)$$

$$\mathcal{E}_K(h, g) := - \int h L_K g \, d\beta = \frac{1}{2} \iint \bar{\nabla} h(x, y) \bar{\nabla} g(x, y) J_\beta(dx, dy), \quad (5)$$

where $\bar{\nabla} h(x, y) = h(y) - h(x)$. Singular kernels are interpreted through the closed Dirichlet form or through principal values.

For $\alpha = r\beta$, with $r \geq 0$ and $\int r \, d\beta = 1$, set

$$\mathcal{H}_\beta(r) := \text{KL}(\alpha \mid \beta) = \int r \log r \, d\beta. \quad (6)$$

The nonlocal Fisher information associated with K is

$$\begin{aligned} \mathcal{I}_K(r) &:= \mathcal{E}_K(r, \log r) \\ &= \frac{1}{2} \iint (r(y) - r(x)) (\log r(y) - \log r(x)) J_\beta(dx, dy). \end{aligned} \quad (7)$$

The integrand is nonnegative. Equivalently, using the logarithmic mean

$$\theta(a, b) = \frac{a - b}{\log a - \log b},$$

one can write

$$\mathcal{I}_K(r) = \frac{1}{2} \iint \left| \bar{\nabla} \log r(x, y) \right|^2 \theta(r(x), r(y)) J_\beta(dx, dy). \quad (8)$$

The logarithmic mean is what converts the entropy gradient into the linear jump evolution. To make the metric statement precise, however, one first needs the underlying dynamic action.

2.2 Nonlocal action and path metric

Let r be a density with respect to β and let $j(x, y) = -j(y, x)$ be an antisymmetric flux density with respect to J_β . Its logarithmic-mean action is

$$\mathcal{A}_K(r, j) := \frac{1}{2} \iint \frac{j(x, y)^2}{\theta(r(x), r(y))} J_\beta(\mathrm{d}x, \mathrm{d}y), \quad (9)$$

with the usual lower-semicontinuous convention at zero. A curve $t \mapsto r_t$ and a flux j_t solve the nonlocal continuity equation when

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi r_t \, \mathrm{d}\beta = \frac{1}{2} \iint \bar{\nabla} \varphi(x, y) j_t(x, y) J_\beta(\mathrm{d}x, \mathrm{d}y) \quad (10)$$

for every admissible test function φ . The associated extended path distance is

$$\mathcal{W}_K^2(r_0\beta, r_1\beta) := \inf_{(r_t, j_t)} \int_0^1 \mathcal{A}_K(r_t, j_t) \, \mathrm{d}t, \quad (11)$$

where the infimum is over solutions of (10) with the prescribed endpoints. For singular kernels, the rigorous relaxation allows measure-valued fluxes rather than only densities j_t .

This construction is the jump analogue of Benamou–Brenier transport. Its metric and geodesic properties have been established under broad hypotheses by Erbar [5], Slepčev and Warren [11]; the distance may be infinite between distinct finite-action components.

2.3 Semigroup dissipation versus metric slope

The reversible semigroup $P_t = e^{tL_K}$ evolves densities according to

$$\partial_t r_t = L_K r_t, \quad \alpha_t = r_t \beta. \quad (12)$$

Whenever differentiation is justified, reversibility alone gives the entropy-dissipation identity

$$-\frac{\mathrm{d}}{\mathrm{d}t} \mathcal{H}_\beta(r_t) = \mathcal{I}_K(r_t). \quad (13)$$

Indeed, differentiating (6) yields $\int (1 + \log r_t) L_K r_t \, \mathrm{d}\beta = -\mathcal{E}_K(r_t, \log r_t)$.

The stronger metric statement uses the flux

$$j_t(x, y) = -\theta(r_t(x), r_t(y)) \bar{\nabla} \log r_t(x, y) = r_t(x) - r_t(y), \quad (14)$$

which turns (10) into (12) and turns (9) into $\mathcal{I}_K(r_t)$. In the settings where the corresponding upper-gradient theory is valid, one obtains

$$|\partial \mathcal{H}_\beta|_{\mathcal{W}_K}^2(r) = \mathcal{I}_K(r) \quad (15)$$

and $P_t r$ is the $\mathcal{W}_K$ -gradient flow of entropy. This is rigorous for the translation-invariant framework of Erbar [5] and for the standard finite-state theory [6]; extending the identification to a given singular, target-dependent kernel requires checking the metric-domain and lower-semicontinuity hypotheses. Below, semigroup statements use only (13); every claim phrased as a metric PL statement additionally assumes (15).

3 MLSI, entropy decay, and metric PL

3.1 MLSI and exponential entropy decay

Definition 3.1 (Modified log-Sobolev inequality). The pair (β, K) satisfies the modified log-Sobolev inequality with constant $\kappa > 0$ if

$$\mathcal{I}_K(r) \geq 2\kappa \mathcal{H}_\beta(r) \quad (16)$$

for every probability density r in the form domain. The optimal constant is

$$\kappa_{\text{MLSI}}(\beta, K) = \frac{1}{2} \inf_{\substack{r \geq 0, \int r d\beta = 1 \\ \mathcal{H}_\beta(r) > 0}} \frac{\mathcal{I}_K(r)}{\mathcal{H}_\beta(r)}. \quad (17)$$

When (15) holds, this is also the optimal metric PL constant of $\mathcal{H}_\beta$ for $\mathcal{W}_K$.

The adjective “modified” matters. For local diffusions, the chain rule identifies the entropy production with four times the Dirichlet energy of $\sqrt{r}$. For jump generators there is no exact chain rule, and $\mathcal{E}_K(r, \log r)$ is not the same object as $4\mathcal{E}_K(\sqrt{r}, \sqrt{r})$. Modified log-Sobolev inequalities are therefore the natural entropy-production inequalities for jump processes; see Adamczak et al. [1], Bobkov and Tetali [3].

Theorem 3.2 (MLSI is equivalent to exponential entropy decay). *Assume that the reversible semigroup $P_t = e^{tL_K}$ satisfies the entropy dissipation identity (13). Then the following are equivalent.*

- (i) *The modified log-Sobolev inequality (16) holds with constant κ .*
- (ii) *For every probability density r and every $t \geq 0$,*

$$\mathcal{H}_\beta(P_t r) \leq e^{-2\kappa t} \mathcal{H}_\beta(r). \quad (18)$$

If, in addition, (15) holds, $P_t r$ is the $\mathcal{W}_K$ -gradient flow, and $P_t r \rightarrow 1$, then

$$\mathcal{W}_K^2(\alpha, \beta) \leq \frac{2}{\kappa} \text{KL}(\alpha \mid \beta) \quad (19)$$

for every α in the same finite-distance component as β .

Proof. Along the semigroup, (13) and (16) give

$$\frac{d}{dt} \mathcal{H}_\beta(P_t r) \leq -2\kappa \mathcal{H}_\beta(P_t r),$$

and Gronwall’s lemma proves (18). Conversely, differentiate (18) at $t = 0^+$. The dissipation identity $-\frac{d}{dt}|_{t=0^+} \mathcal{H}_\beta(P_t r) = \mathcal{I}_K(r)$ gives the reverse implication.

For the last assertion, the standard PL tail-length argument in the metric space $(\mathcal{P}(\mathcal{X}), \mathcal{W}_K)$ gives

$$\mathcal{W}_K(\alpha, \beta) \leq \int_0^\infty |\dot{\alpha}_t|_{\mathcal{W}_K} dt \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{H}_\beta(r)}.$$

Squaring yields (19). □

Thus the requested “fractional log-Sobolev inequality” is exactly MLSI at the semigroup level and, under (15), exactly PL in the reversible nonlocal Wasserstein geometry. What changes with the kernel is the entropy production $\mathcal{I}_K$.

3.2 Linearization: a spectral gap is necessary

Let h be bounded, centered under β , and set $r_\varepsilon = 1 + \varepsilon h$. As $\varepsilon \rightarrow 0$,

$$\mathcal{H}_\beta(r_\varepsilon) = \frac{\varepsilon^2}{2} \int h^2 d\beta + O(\varepsilon^3), \quad \mathcal{I}_K(r_\varepsilon) = \varepsilon^2 \mathcal{E}_K(h, h) + O(\varepsilon^3). \quad (20)$$

Consequently:

Proposition 3.3 (PL implies the jump Poincaré inequality). *If (16) holds with constant κ , then*

$$\mathcal{E}_K(h, h) \geq \kappa \operatorname{Var}_\beta(h) \quad (21)$$

for every h in the Dirichlet-form domain.

Proof. Insert (20) into (16), divide by ε^2 , and let $\varepsilon \rightarrow 0$. $\square$

The converse is false without additional assumptions: a spectral gap is an L^2 statement near equilibrium, whereas MLSI controls arbitrary densities, including highly concentrated ones.

3.3 Kurdyka–Łojasiewicz profiles

The power law is only one possible weakening of PL. Let $\omega : [0, M) \rightarrow [0, \infty)$ be continuous at zero, increasing and concave, with $\omega(0) = 0$ and $\omega \in C^1((0, M))$. A nonlocal KL inequality on the entropy sublevel $\{0 < \mathcal{H}_\beta < M\}$ is

$$\omega'(\mathcal{H}_\beta(r)) \sqrt{\mathcal{I}_K(r)} \geq 1. \quad (22)$$

Along the semigroup, this is an entropy–entropy-production inequality. If the metric slope identity (15) also holds, it is the usual metric Kurdyka–Łojasiewicz inequality and bounds the remaining metric length by

$$\int_t^\infty |\dot{\alpha}_s|_{\mathcal{W}_K} ds \leq \omega(\mathcal{H}_\beta(r_t)). \quad (23)$$

Indeed, combine $-\dot{\mathcal{H}}_\beta = \mathcal{I}_K$ with $|\dot{\alpha}_t|_{\mathcal{W}_K} = \sqrt{\mathcal{I}_K(r_t)}$ and integrate (22).

The most transparent family is the power profile. Taking $\omega(s) = s^{1-\vartheta}/[c(1-\vartheta)]$ in (22) gives

$$\sqrt{\mathcal{I}_K(r)} \geq c \mathcal{H}_\beta(r)^\vartheta, \quad \vartheta \in [0, 1), \quad (24)$$

on a specified entropy sublevel set or regularity class. The case $\vartheta = 1/2$ is MLSI, and also metric PL when (15) holds, with $c = \sqrt{2\kappa}$.

Proposition 3.4 (Entropy rates under nonlocal power-KL). *Assume (24) along the entropy flow. If $\vartheta > 1/2$, then*

$$\mathcal{H}_\beta(r_t) \leq \left(\mathcal{H}_\beta(r_0)^{1-2\vartheta} + c^2(2\vartheta-1)t \right)^{-1/(2\vartheta-1)}. \quad (25)$$

If $\vartheta = 1/2$, the decay is exponential with rate c^2 . Moreover, a global inequality with $\vartheta < 1/2$ is impossible near a nondegenerate equilibrium whenever the tangent Dirichlet form is nonzero.

Proof. The dissipation identity gives $\frac{d}{dt} \mathcal{H}_\beta(r_t) = -\mathcal{I}_K(r_t) \leq -c^2 \mathcal{H}_\beta(r_t)^{2\vartheta}$, whose integration yields (25). For the last assertion, insert $r_\varepsilon = 1 + \varepsilon h$ into (24). By (20), its two sides scale respectively as $|\varepsilon|$ and $|\varepsilon|^{2\vartheta}$. If $\vartheta < 1/2$, the right-hand side dominates as $\varepsilon \rightarrow 0$, which is impossible for a fixed $c > 0$. $\square$

Power-KL inequalities are useful for obtaining polynomial rates, but they do not remove the need to specify the kernel. They are entropy–entropy production inequalities for the pair (β, K) just as MLSI is.

4 Why the target measure alone cannot decide PL

4.1 Scaling, connectivity, and bottlenecks

Replacing K by δK multiplies L_K , $\mathcal{E}_K$, and $\mathcal{I}_K$ by δ , and divides $\mathcal{W}_K$ by $\sqrt{\delta}$. Thus the optimal MLSI constant scales as

$$\kappa(\beta, \delta K) = \delta \kappa(\beta, K). \quad (26)$$

No quantitative MLSI constant can therefore follow from β alone. If K does not connect two positive- β sets, a density that is constant on each component has positive entropy and zero dissipation, so even the existence of a positive constant fails.

The obstruction can be quantified exactly by testing two-level densities. For a measurable set A , write $p = \beta(A)$ and define its edge conductance

$$Q_K(A, A^c) := J_\beta(A \times A^c). \quad (27)$$

Proposition 4.1 (Exact two-level bottleneck test). *Let $0 < p < 1$, $q \in (0, 1)$, and*

$$r_{A,q} = \frac{q}{p} \mathbf{1}_A + \frac{1-q}{1-p} \mathbf{1}_{A^c}. \quad (28)$$

Then

$$\mathcal{H}_\beta(r_{A,q}) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}, \quad (29)$$

$$\mathcal{I}_K(r_{A,q}) = Q_K(A, A^c) \left(\frac{q}{p} - \frac{1-q}{1-p} \right) \log \frac{q(1-p)}{p(1-q)}. \quad (30)$$

Consequently (17) is at most

$$\frac{Q_K(A, A^c)}{2\mathcal{H}_\beta(r_{A,q})} \left(\frac{q}{p} - \frac{1-q}{1-p} \right) \log \frac{q(1-p)}{p(1-q)}. \quad (31)$$

As $q \rightarrow p$, this upper bound tends to $Q_K(A, A^c)/[p(1-p)]$, the Rayleigh quotient of $\mathbf{1}_A$.

Proof. The density (28) assigns masses q and $1-q$ to A and A^c , so direct integration gives (29). In (7), pairs lying in the same set contribute zero. Symmetry of J_β cancels the factor $1/2$ between the two orientations of $A \times A^c$ and gives (30). Inserting this test density into (17) proves (31). Finally, writing $q = p + \delta$ gives

$$\begin{aligned} \mathcal{H}_\beta(r_{A,p+\delta}) &= \frac{\delta^2}{2p(1-p)} + o(\delta^2), \\ \mathcal{I}_K(r_{A,p+\delta}) &= \frac{Q_K(A, A^c)\delta^2}{p^2(1-p)^2} + o(\delta^2), \end{aligned}$$

which proves the limit. □

This proposition separates two obstructions. Taking $q \rightarrow p$ detects the linearized spectral gap, while taking q far from p probes the genuinely nonlinear entropy cost of concentrating mass behind a bottleneck. Irreducibility by itself does not rule either out. The following example fixes the target to be Gaussian and uses an everywhere-positive kernel.

Proposition 4.2 (A Gaussian target with no MLSI). *Let $\beta = \mathcal{N}(0, I_d)$ and let $a : \mathbb{R}^d \rightarrow (0, 1]$ satisfy $a(x) \rightarrow 0$ as $|x| \rightarrow \infty$, for instance $a(x) = e^{-|x|^2}$. Define*

$$K_a(x, dy) = a(x)a(y)\beta(dy). \quad (32)$$

Then K_a is everywhere positive and reversible with respect to β , but its Dirichlet form has no positive spectral gap. Consequently the pair has no positive MLSI constant and, whenever (15) holds, $\text{KL}(\cdot \mid \beta)$ satisfies no global metric PL inequality. For the specific choice $a(x) = e^{-|x|^2}$, it satisfies no power-KL inequality (24) with any $\vartheta \in [0, 1)$ on any nontrivial entropy sublevel $\{r : \mathcal{H}_\beta(r) \leq M\}$, $M > 0$.

Proof. The edge measure $a(x)a(y)\beta(\mathrm{d}x)\beta(\mathrm{d}y)$ is symmetric, hence the kernel is reversible. Let $A_R = \{x : |x| \geq R\}$ and $p_R = \beta(A_R)$. Its conductance factorizes as

$$Q_{K_a}(A_R, A_R^c) = \left(\int_{A_R} a \, \mathrm{d}\beta \right) \left(\int_{A_R^c} a \, \mathrm{d}\beta \right) \leq p_R \sup_{A_R} a.$$

The $q \rightarrow p_R$ limit in Proposition 4.1 is at most $\sup_{A_R} a/(1 - p_R)$ and tends to zero. Hence the spectral gap and MLSI constants both vanish.

For the stronger assertion, fix $h > 0$, set $L_R = \log(1/p_R)$ and $q_R = h/L_R$, and consider the two-level density

$$r_R = r_{A_R, q_R}.$$

As $R \rightarrow \infty$, direct substitution gives $\mathcal{H}_\beta(r_R) \rightarrow h$. On the other hand, the factorization of the edge measure in (32) gives

$$\begin{aligned} \mathcal{I}_{K_a}(r_R) &= Q_{K_a}(A_R, A_R^c) \left(\frac{q_R}{p_R} - \frac{1 - q_R}{1 - p_R} \right) \\ &\quad \times \log \left(\frac{q_R(1 - p_R)}{p_R(1 - q_R)} \right). \end{aligned}$$

For $a(x) = e^{-|x|^2}$, this is $O(e^{-R^2} q_R L_R) = O(e^{-R^2})$, hence tends to zero. Thus a positive power-KL constant is impossible on $\{r : \mathcal{H}_\beta(r) \leq M\}$: choose any $h \in (0, M)$ and take R sufficiently large. $\square$

This example is the clean reason that “ β is Gaussian” or “ β is strongly log-concave” cannot by itself answer the question. One must specify how the process is allowed to jump.

5 Verifiable sufficient conditions

5.1 Positive entropic curvature

The geometric sufficient condition is the exact analogue of the local Wasserstein argument.

Proposition 5.1 (Positive nonlocal entropy curvature implies PL). *Assume that $\mathcal{H}_\beta$ is λ -geodesically convex in $(\mathcal{P}(\mathcal{X}), \mathcal{W}_K)$ for some $\lambda > 0$, and that its unique minimizer is β . Then (16) holds with $\kappa = \lambda$.*

Proof. The standard metric slope estimate for a λ -geodesically convex functional gives $|\partial \mathcal{H}_\beta|_{\mathcal{W}_K}^2 \geq 2\lambda \mathcal{H}_\beta$. Use the slope identity (15). $\square$

This is the entropic Ricci-curvature route developed for Markov chains by Erbar and Maas [6]. For the translation-invariant Lévy kernels treated by Erbar [5], the proved statement is only 0-geodesic convexity. It gives contractivity but no positive PL constant; strict curvature for concrete singular nonhomogeneous kernels is much harder to verify.

5.2 Gross log-Sobolev implies modified log-Sobolev

Suppose the jump Dirichlet form satisfies the Gross inequality

$$\text{Ent}_\beta(h^2) \leq \frac{2}{\lambda_G} \mathcal{E}_K(h, h). \quad (33)$$

This is not the same inequality as (16), but it is a useful strong sufficient condition.

Proposition 5.2 (Gross LSI implies MLSI). *If (33) holds, then $\mathcal{I}_K(r) \geq 2\lambda_G \mathcal{H}_\beta(r)$. Thus the MLSI constant is at least λ_G ; under (15), so is the metric PL constant.*

Proof. For $a, b > 0$,

$$(a - b)(\log a - \log b) \geq 4(\sqrt{a} - \sqrt{b})^2. \quad (34)$$

Integrating this inequality against $J_\beta/2$ gives $\mathcal{I}_K(r) \geq 4\mathcal{E}_K(\sqrt{r}, \sqrt{r})$. Apply (33) to $h = \sqrt{r}$, for which $\text{Ent}_\beta(h^2) = \mathcal{H}_\beta(r)$. $\square$

The inequality (34) replaces the exact diffusion chain rule. It is also the bridge allowing the substantial literature on Poincaré, super-Poincaré, and Gross log-Sobolev inequalities for stable-like Dirichlet forms to provide PL results for the nonlocal Wasserstein entropy flow.

5.3 A Doeblin or refresh minorization

The simplest nonlocal kernel is the refresh kernel $K_{\text{ref}}(x, dy) = \beta(dy)$. It forgets the current state and draws a new state from equilibrium. Its entropy production is explicit:

$$\begin{aligned} \mathcal{I}_{\text{ref}}(r) &= \frac{1}{2} \iint (r(x) - r(y))(\log r(x) - \log r(y)) \, d\beta(x) \, d\beta(y) \\ &= \mathcal{H}_\beta(r) - \int \log r \, d\beta = \text{KL}(\alpha \mid \beta) + \text{KL}(\beta \mid \alpha) \geq \mathcal{H}_\beta(r). \end{aligned} \quad (35)$$

Proposition 5.3 (Doeblin minorization implies MLSI). *Assume the symmetric edge measure satisfies*

$$J_\beta(dx, dy) \geq c\beta(dx)\beta(dy) \quad (36)$$

for some $c > 0$. Then $\mathcal{I}_K(r) \geq c\mathcal{H}_\beta(r)$, so (16) holds with $\kappa = c/2$.

Proof. The integrand in (7) is nonnegative. Compare the edge measure with (35) and use Jensen's inequality $\int \log r \, d\beta \leq \log \int r \, d\beta = 0$. $\square$

This condition is strong, but it makes the roles of the target and the kernel transparent: β supplies the refresh law, while c quantifies the jump mechanism's access to it.

5.4 Spectral gap plus bounded densities

A spectral gap becomes sufficient after restricting the density ratio.

Proposition 5.4 (MLSI on an L^∞ density class). *Assume the Poincaré inequality $\mathcal{E}_K(h, h) \geq \lambda_P \text{Var}_\beta(h)$ and let r be a probability density satisfying $0 \leq r \leq M$. Then*

$$\mathcal{I}_K(r) \geq \frac{\lambda_P}{M} \mathcal{H}_\beta(r). \quad (37)$$

Thus MLSI holds on this class with constant $\kappa = \lambda_P/(2M)$; it is a restricted metric PL estimate under (15).

Proof. For $0 < a, b \leq M$, the mean-value theorem gives $(a - b)(\log a - \log b) \geq (a - b)^2/M$. Hence $\mathcal{I}_K(r) \geq \mathcal{E}_K(r, r)/M$. Moreover, $\mathcal{H}_\beta(r) \leq \int (r - 1)^2 \, d\beta$ because $u \log u - u + 1 \leq (u - 1)^2$. The Poincaré inequality concludes the proof. $\square$

This estimate can yield *eventual* exponential entropy decay when the jump semigroup regularizes arbitrary initial data into a bounded density class. The constant then depends on the regularization bound and is not a global MLSI constant uniform over all initial densities.

5.5 Weak log-Sobolev inequalities give KL profiles

When a global MLSI fails, a weak log-Sobolev inequality can still quantify the decay. Assume that a decreasing profile $B_{\text{WL}} : (0, \infty) \rightarrow (0, \infty)$ satisfies

$$\text{Ent}_\beta(h^2) \leq B_{\text{WL}}(s) \mathcal{E}_K(h, h) + s \text{Osc}(h)^2, \quad s > 0, \quad (38)$$

where $\text{Osc}(h) = \text{ess sup } h - \text{ess inf } h$. Such inequalities and their relation to entropy decay are developed by Cattiaux et al. [4].

Proposition 5.5 (Weak LSI yields a restricted nonlocal KL inequality). *Assume (38). Let $r_t = P_t r_0$, with r_0 bounded, and set $\Omega = \text{Osc}(\sqrt{r_0})$. Whenever $\mathcal{H}_\beta(r_t) > 0$ and $\Omega > 0$,*

$$\mathcal{I}_K(r_t) \geq \frac{2\mathcal{H}_\beta(r_t)}{B_{\text{WL}}(\mathcal{H}_\beta(r_t)/(2\Omega^2))}. \quad (39)$$

In particular, if $B_{\text{WL}}(s) \leq Cs^{-a}$ for some $a > 0$, then

$$\mathcal{I}_K(r_t) \geq \frac{2^{1-a}}{C\Omega^{2a}} \mathcal{H}_\beta(r_t)^{1+a}, \quad (40)$$

$$\mathcal{H}_\beta(r_t) \leq \left[\mathcal{H}_\beta(r_0)^{-a} + \frac{a2^{1-a}}{C\Omega^{2a}} t \right]^{-1/a}. \quad (41)$$

If $0 < a < 1$, this is a genuine power-KL inequality with exponent $\vartheta = (1+a)/2 \in (1/2, 1)$ on the class with fixed oscillation bound. Under (15), it controls the metric tail length through (23). For $a \geq 1$, the same entropy-production bound and polynomial rate remain valid, but the formal power exponent is at least one and no finite desingularizer ω at zero follows from this argument.

Proof. Apply (38) to $h = \sqrt{r_t}$ and use (34) to obtain

$$\mathcal{H}_\beta(r_t) \leq \frac{1}{4} B_{\text{WL}}(s) \mathcal{I}_K(r_t) + s \text{Osc}(\sqrt{r_t})^2.$$

The Markov property keeps $P_t r_0$ between the essential infimum and supremum of r_0 , hence $\text{Osc}(\sqrt{r_t}) \leq \Omega$. Choosing $s = \mathcal{H}_\beta(r_t)/(2\Omega^2)$ and rearranging proves (39). The polynomial bound on B_{WL} gives (40); integrating $-\dot{\mathcal{H}}_\beta = \mathcal{I}_K$ then yields (41). $\square$

The dependence on Ω is essential: weak functional inequalities do not produce a uniform global KL constant over arbitrarily concentrated initial densities. They instead give a hierarchy of rates on controlled classes, which is precisely the regime missed by a binary PL/no-PL diagnosis.

5.6 Comparison of kernels

If two reversible edge measures satisfy $J_1 \geq cJ_0$, then $\mathcal{I}_{K_1}(r) \geq c\mathcal{I}_{K_0}(r)$. Consequently any PL, power-KL, Gross log-Sobolev, or Poincaré estimate for K_0 transfers to K_1 with the corresponding factor. This comparison principle is often more practical than computing nonlocal curvature.

6 Why the pure fractional Laplacian is not a confined entropy flow

6.1 Translation invariance fixes the reversible measure

Consider a symmetric translation-invariant jump kernel

$$K(x, dy) = q(y - x) dy, \quad q(z) = q(-z). \quad (42)$$

The fractional Laplacian corresponds to $q(z) = c_{d,s} |z|^{-d-s}$, $0 < s < 2$.

Proposition 6.1 (No nonconstant reversible target). *Let $\beta(dx) = p(x) dx$ with $p > 0$ a.e. Assume that the support of q generates $\mathbb{R}^d$ as an additive group. If (42) is reversible with respect to β , then p is constant a.e. In particular, on $\mathbb{R}^d$ no probability measure, Gaussian or otherwise, is reversible for the pure fractional Laplacian.*

Proof. Detailed balance reads $p(x)q(y-x) = p(y)q(x-y)$. By symmetry of q , this gives $p(x) = p(y)$ whenever $q(y-x) > 0$. Connectivity under sums of admissible increments makes p constant. $\square$

For the pure fractional heat equation

$$\partial_t \rho_t = -(-\Delta)^{s/2} \rho_t, \quad (43)$$

the reference measure is Lebesgue measure. There is no invariant probability and no finite entropy minimizer. Indeed, for $\rho_R(x) = R^{-d} \rho(x/R)$,

$$\int \rho_R \log \rho_R dx = \int \rho \log \rho dx - d \log R \rightarrow -\infty.$$

Therefore a global PL inequality around a probability equilibrium is not the right question for (43). Euclidean fractional log-Sobolev inequalities do exist, but they are scale-sensitive Sobolev inequalities used for smoothing and ultracontractivity, not MLSI around a confining target; see Gentil and Imbert [7], Gressman [8].

6.2 Adding a drift changes the invariant law and may break reversibility

The fractional Fokker–Planck equation

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) - (-\Delta)^{s/2} \rho_t \quad (44)$$

is the forward equation of a Langevin SDE driven by symmetric stable noise. For $s < 2$, its invariant law is generally *not* $Z^{-1} e^{-V} dx$. Even for the quadratic drift $\nabla V(x) = x$, the invariant law is s -stable rather than Gaussian. Moreover this Lévy-driven Ornstein–Uhlenbeck process is generally not reversible when $s < 2$, so its entropy production need not be the metric slope for the symmetric nonlocal Wasserstein distance considered here.

Gentil and Imbert [7] prove modified Φ -entropy inequalities and exponential convergence for such Lévy–Fokker–Planck equations. In the s -stable Ornstein–Uhlenbeck case their entropy estimate has the form $\text{Ent}(r_t) \leq e^{-st} \text{Ent}(r_0)$. This is an important entropy-production result, but for $s < 2$ it should not be silently identified with a PL theorem for a reversible nonlocal Wasserstein metric. The reversible Gaussian model constructed below is a different operator.

7 A canonical positive construction by subordination

7.1 Start from a reversible Langevin semigroup

Let

$$\beta(dx) = Z^{-1} e^{-V(x)} dx$$

and consider the reversible Langevin generator

$$L_\beta h = \Delta h - \nabla V \cdot \nabla h. \quad (45)$$

Write $A = -L_\beta \geq 0$ on $L^2(\beta)$ and $P_t = e^{-tA}$. Assume that β satisfies the ordinary log-Sobolev inequality with constant $\lambda > 0$, equivalently

$$\mathcal{H}_\beta(P_t r) \leq e^{-2\lambda t} \mathcal{H}_\beta(r). \quad (46)$$

The Bakry–Émery criterion gives (46) whenever

$$\nabla^2 V \succeq \lambda I_d. \quad (47)$$

Thus strong log-concavity is a concrete sufficient condition at the local level [2].

7.2 Bochner subordination

Let ϕ be a Bernstein function with $\phi(0) = 0$. There is a convolution semigroup of probability measures $(\eta_t)_{t \geq 0}$ on $[0, \infty)$ such that

$$\int_0^\infty e^{-qs} \eta_t(ds) = e^{-t\phi(q)}. \quad (48)$$

The subordinate Markov semigroup is

$$P_t^\phi r := \int_0^\infty P_s r \eta_t(ds), \quad P_t^\phi = e^{-t\phi(A)}. \quad (49)$$

This is Bochner subordination; see Schilling and Wang [9], Schilling et al. [10].

If

$$\phi(q) = bq + \int_0^\infty (1 - e^{-qt}) \nu_\phi(dt), \quad (50)$$

then

$$-\phi(A)h = bL_\beta h + \int_0^\infty (P_t h - h) \nu_\phi(dt). \quad (51)$$

When $b = 0$, this is a pure-jump generator. If $P_t(x, dy)$ denotes the transition kernel of the Langevin semigroup, its jump kernel is

$$K_\phi(x, dy) = \int_0^\infty P_t(x, dy) \nu_\phi(dt), \quad y \neq x. \quad (52)$$

Since every P_t is reversible with respect to β , so is K_ϕ . The corresponding entropy production is

$$\mathcal{I}_\phi(r) = \langle r, \phi(A) \log r \rangle_{L^2(\beta)} = - \left. \frac{d}{dt} \right|_{t=0^+} \mathcal{H}_\beta(P_t^\phi r). \quad (53)$$

Theorem 7.1 (Subordination transfers MLSI). *Assume (46). Then*

$$\mathcal{H}_\beta(P_t^\phi r) \leq e^{-t\phi(2\lambda)} \mathcal{H}_\beta(r), \quad (54)$$

and

$$\mathcal{I}_\phi(r) \geq \phi(2\lambda) \mathcal{H}_\beta(r). \quad (55)$$

Consequently the subordinate generator's MLSI constant satisfies

$$\kappa_\phi \geq \frac{\phi(2\lambda)}{2}. \quad (56)$$

If $b = 0$ and the logarithmic-mean metric for K_ϕ satisfies (15), the same bound holds for its metric PL constant.

Proof. The map $u \mapsto u \log u$ is convex. Jensen's inequality in (49), followed by (46), yields

$$\begin{aligned} \mathcal{H}_\beta(P_t^\phi r) &\leq \int_0^\infty \mathcal{H}_\beta(P_s r) \eta_t(ds) \\ &\leq \mathcal{H}_\beta(r) \int_0^\infty e^{-2\lambda s} \eta_t(ds) = e^{-t\phi(2\lambda)} \mathcal{H}_\beta(r), \end{aligned}$$

where the last equality is (48). Differentiate at $t = 0^+$ and use (53). The normalization $\mathcal{I}_\phi \geq 2\kappa_\phi \mathcal{H}_\beta$ gives (56); when $b = 0$, this is precisely Definition 3.1 for K_ϕ . $\square$

This argument requires only entropy contraction of the base semigroup, not pointwise curvature. Therefore any target satisfying a classical LSI can be used, even when its LSI is proved by a method other than strong log-concavity.

The lower bound has a useful spectral counterpart. Let λ_{gap} be the optimal Poincaré constant of A .

Proposition 7.2 (Subordinate MLSI bracket). *Under the assumptions of Theorem 7.1, let κ_ϕ denote the optimal constant in $\mathcal{I}_\phi \geq 2\kappa_\phi \mathcal{H}_\beta$. Then*

$$\frac{\phi(2\lambda)}{2} \leq \kappa_\phi \leq \phi(\lambda_{\text{gap}}). \quad (57)$$

When $b = 0$, this is $\kappa_{\text{MLSI}}(\beta, K_\phi)$ as defined in (17); the full bracket also applies when (50) contains a local drift term bA .

Proof. The lower bound is Theorem 7.1. By spectral calculus, the optimal Poincaré constant of $\phi(A)$ is $\phi(\lambda_{\text{gap}})$, since ϕ is increasing and $\phi(0) = 0$. Proposition 3.3 says that an MLSI constant cannot exceed this spectral gap. $\square$

When $b > 0$, the generator (51) combines a local diffusion with jumps, so its gradient structure is not a purely nonlocal transport geometry. The fractional powers $\phi(q) = q^{s/2}$ with $0 < s < 2$ have $b = 0$ and are genuinely pure jump.

8 The Gaussian target

8.1 Fractional powers of the Ornstein–Uhlenbeck generator

Take $\beta = \gamma_d = \mathcal{N}(0, I_d)$ and

$$L_{\text{OU}} = \Delta - x \cdot \nabla, \quad A_{\text{OU}} = -L_{\text{OU}}. \quad (58)$$

The Gaussian log-Sobolev constant is $\lambda = 1$. For a fractional order $0 < s < 2$, choose

$$\phi(q) = q^{s/2}, \quad L_s^{\text{OU}} = -A_{\text{OU}}^{s/2}. \quad (59)$$

This is the fractional power of the *Ornstein–Uhlenbeck generator*, not the operator obtained by replacing Δ with $-(-\Delta)^{s/2}$ in (58).

The Mehler kernel is

$$P_t^{\text{OU}}(x, dy) = \mathcal{N}\left(e^{-t}x, (1 - e^{-2t})I_d\right)(dy). \quad (60)$$

Since

$$q^a = \frac{a}{\Gamma(1-a)} \int_0^\infty (1 - e^{-qt}) \frac{dt}{t^{1+a}}, \quad a = s/2 \in (0, 1),$$

the explicit Gaussian-reversible jump kernel is

$$K_s^{\text{OU}}(x, dy) = \frac{s/2}{\Gamma(1-s/2)} \int_0^\infty \mathcal{N}\left(e^{-t}x, (1 - e^{-2t})I_d\right)(dy) \frac{dt}{t^{1+s/2}}, \quad y \neq x. \quad (61)$$

At short scales this is a singular fractional-type jump mechanism; globally it is tilted by the Ornstein–Uhlenbeck dynamics so that the Gaussian remains reversible.

Corollary 8.1 (Gaussian fractional-OU MLSI). *For the generator (59),*

$$\mathcal{I}_s^{\text{OU}}(r) \geq 2^{s/2} \mathcal{H}_{\gamma_d}(r). \quad (62)$$

Hence its MLSI constant satisfies

$$\kappa_s^{\text{OU}} \geq 2^{s/2-1}. \quad (63)$$

Moreover the optimal constant obeys

$$2^{s/2-1} \leq \kappa_{s,\text{opt}}^{\text{OU}} \leq 1. \quad (64)$$

Proof. The lower bound is Theorem 7.1 with $\lambda = 1$ and $\phi(q) = q^{s/2}$. For the upper bound, use the Gaussian translation density

$$r_m(x) = \exp(mx_1 - m^2/2), \quad r_m \gamma_d = \mathcal{N}(me_1, I_d).$$

One has $\mathcal{H}_{\gamma_d}(r_m) = m^2/2$. Since constants are annihilated by $A_{\text{OU}}^{s/2}$ and x_1 is an eigenfunction with eigenvalue 1,

$$\mathcal{I}_s^{\text{OU}}(r_m) = \langle r_m, A_{\text{OU}}^{s/2} \log r_m \rangle_{L^2(\gamma_d)} = m \int x_1 r_m \, d\gamma_d = m^2.$$

Thus $\mathcal{I}_s^{\text{OU}}(r_m)/(2\mathcal{H}_{\gamma_d}(r_m)) = 1$, which forces $\kappa_{s,\text{opt}}^{\text{OU}} \leq 1$. $\square$

The bound from subordination is sharp at $s = 2$, where the classical Gaussian LSI gives $\kappa = 1$. For $0 < s < 2$, determining the sharp MLSI constant of $A_{\text{OU}}^{s/2}$ remains a separate problem; the simple Jensen argument gives a dimension-free lower bound but need not be sharp.

8.2 Direct stable-like kernels for stretched-exponential targets

There is another natural construction, closer pointwise to the fractional Laplacian. Let $p = d\beta/dx$ and define the symmetric edge density

$$j_{s,\beta}(x, y) = \frac{c_{d,s}}{2|x-y|^{d+s}}(p(x) + p(y)). \quad (65)$$

Then

$$K_{s,\beta}(x, dy) = \frac{j_{s,\beta}(x, y)}{p(x)} \, dy = \frac{c_{d,s}}{2|x-y|^{d+s}} \left(1 + \frac{p(y)}{p(x)}\right) \, dy \quad (66)$$

is reversible with respect to β . Its Dirichlet form agrees, up to the factor $c_{d,s}/2$, with the stable-like form studied by Wang and Wang [12]:

$$\iint \frac{|h(x) - h(y)|^2}{|x-y|^{d+s}} \, dy \, e^{-V(x)} \, dx.$$

Their estimates apply to a family substantially larger than the Gaussian.

Proposition 8.2 (MLSI for stable-like stretched-exponential targets). *Fix $0 < s < 2$ and $q > 0$, and let*

$$\beta_q(dx) = Z_q^{-1} \exp\left(-(1+|x|^2)^q\right) \, dx. \quad (67)$$

For the reversible kernel (66), there is a constant $\kappa_{d,s,q} > 0$ such that

$$\mathcal{I}_{K_{s,\beta_q}}(r) \geq 2\kappa_{d,s,q} \mathcal{H}_{\beta_q}(r). \quad (68)$$

If the metric slope identity (15) is available for this singular target-dependent kernel, the same estimate is the corresponding nonlocal Wasserstein-PL inequality.

Proof. Corollary 1.5(2) of Wang and Wang [12] gives a super-Poincaré inequality for the stable-like form and every $q > 0$, with a profile that grows polynomially, up to a logarithmic factor, as its scale parameter tends to zero. Such a profile is bounded by $\exp(C(1+r^{-1}))$ after adjusting C . The standard equivalence between this super-Poincaré bound and a Gross log-Sobolev inequality therefore gives (33) for some $\lambda_G > 0$; see also the equivalence recalled in the introduction of the cited paper. Proposition 5.2 then proves (68), after absorbing the fixed normalization of (65) into the constant. $\square$

For $q \geq 1/2$, the potential $(1 + |x|^2)^q$ is convex: its radial Hessian eigenvalue is $2q(1 + |x|^2)^{q-2}[1 + (2q - 1)|x|^2]$, and its tangential eigenvalues are positive. Hence (67) is log-concave. For $1/2 \leq q < 1$, its tails are slower than Gaussian and it does not satisfy the ordinary Gross LSI for the local gradient Dirichlet form, since that inequality implies Gaussian concentration [2]. Proposition 8.2 shows that the stable-like jump form nevertheless satisfies a Gross LSI and hence MLSI. Long jumps can thus restore entropy coercivity that is absent from the local Langevin geometry. For $q \geq 1$, the potential is in fact strongly convex, so both this direct stable-like construction and the subordinate-Langevin construction apply. The case $q = 1$ is Gaussian up to a harmless rescaling.

For a Gaussian target, this construction and fractional OU subordination are different. Both are Gaussian-reversible and nonlocal, but their edge kernels and spectra differ. It is therefore misleading to ask for “the” fractional Gaussian MLSI, or metric PL, constant without fixing the generator.

9 Assessment for log-concave targets

The preceding analysis can be summarized as follows.

Assumption/model	Conclusion	Reason
β log-concave, arbitrary reversible K	No general MLSI conclusion	Kernel scaling and tail bottlenecks are independent of log-concavity.
$\nabla^2 V \succeq \lambda I$, subordinate Langevin kernel	MLSI with $\kappa \geq \phi(2\lambda)/2$	Bakry–Émery LSI followed by Theorem 7.1.
β satisfies an LSI by any method, subordinate kernel	Same MLSI conclusion	Only entropy contraction of the base semigroup is used.
β merely log-concave on unbounded $\mathbb{R}^d$	No uniform positive constant from convexity alone	Zero curvature gives no LSI rate; exponential-tail examples such as the Laplace law do not satisfy the classical Gaussian-type LSI.
$\beta = \gamma_d$, fractional OU power	$2^{s/2-1} \leq \kappa_{\text{opt}} \leq 1$	Explicit subordinate Mehler kernel and Gaussian LSI.
$\beta = \gamma_d$, symmetrized stable-like kernel	Some $\kappa_{d,s} > 0$	Stable-like super-Poincaré/Gross-LSI theory and Proposition 5.2.
$\beta = \beta_q$ in (67), stable-like kernel	Some $\kappa_{d,s,q} > 0$ for every $q > 0$	Proposition 8.2; for $1/2 \leq q < 1$ this includes log-concave targets without a local Gross LSI.
K satisfies only a weak LSI	Restricted power-KL and polynomial decay	Proposition 5.5, with constants depending on the initial oscillation class.
$\beta = \gamma_d$, tail-degenerate kernel (32)	No MLSI or global power-KL	Tail densities have fixed entropy and vanishing dissipation.

The strongest robust statement is therefore not “log-concavity implies a fractional log-Sobolev inequality.” It is:

Choose a reversible nonlocal generator together with its target. Prove MLSI for that pair, either directly, by curvature, by comparison, through a Gross LSI, or by

subordination of a semigroup whose entropy contraction is already known. To call the result a metric PL theorem, separately verify that the chosen nonlocal action identifies $\mathcal{I}_K$ with the squared entropy slope.

Strong log-concavity is especially useful because it supplies the known base entropy contraction needed by subordination. Mere log-concavity can still be enough for particular targets and kernels, but not by a target-only theorem of the form suggested in the question.

10 A practical theorem template for the book

The following compact statement captures the part that can be integrated safely into the treatment of nonlocal Wasserstein flows.

Theorem 10.1 (Nonlocal entropy-dissipation and PL dictionary). *Let K be reversible with respect to a probability measure β and assume the entropy-dissipation identity (13). For $r = d\alpha/d\beta$, the following statements are equivalent:*

- (i) (β, K) satisfies the modified log-Sobolev inequality $\mathcal{E}_K(r, \log r) \geq 2\kappa \text{Ent}_\beta(r)$;
- (ii) the reversible jump semigroup satisfies $\text{KL}((P_t r)\beta \mid \beta) \leq e^{-2\kappa t} \text{KL}(r\beta \mid \beta)$.

If, moreover, the logarithmic-mean distance (11) satisfies (15), these are equivalent to the $\mathcal{W}_K$ -PL inequality for $f(\alpha) = \text{KL}(\alpha \mid \beta)$. They then imply $\mathcal{W}_K^2(\alpha, \beta) \leq 2\text{KL}(\alpha \mid \beta)/\kappa$ on the finite-distance component of β . In all cases a necessary condition is the spectral-gap inequality $\mathcal{E}_K(h, h) \geq \kappa \text{Var}_\beta(h)$.

The subordinate Langevin theorem can then be presented as the main positive example: strong log-concavity of the target implies a local LSI, and subordination converts it into a reversible fractional jump flow with an explicit MLSI lower bound and, once (15) is verified, the same metric PL lower bound.

11 Open questions and research directions

1. **Sharp Gaussian constant.** Determine the exact MLSI constant of $(-L_{\text{OU}})^{s/2}$. This note gives $2^{s/2-1} \leq \kappa_{\text{opt}} \leq 1$.
2. **Positive nonlocal curvature.** Find checkable assumptions on a nonhomogeneous singular kernel implying positive geodesic convexity of entropy in $\mathcal{W}_K$. Translation-invariant kernels give only zero curvature in the available theory.
3. **Canonical kernels for a prescribed target.** Compare subordinate Langevin, Metropolis-adjusted stable, ground-state transformed, and symmetrized stable-like kernels. They share the target but need not share MLSI constants or even the same finite-distance components.
4. **Mere log-concavity.** Identify additional kernel lower bounds or Lyapunov conditions under which a log-concave target with zero curvature satisfies MLSI. A dimension-free statement cannot depend on the target alone.
5. **Sharp KL profiles.** Proposition 5.5 converts weak log-Sobolev profiles into power-KL bounds, but generally not sharply. Determine the optimal desingularizer ω for concrete tail-degenerate kernels and decide when the resulting entropy rates are attained.
6. **Metric rigor for singular target-dependent kernels.** For kernels such as (61) and (66), verify in one common framework the lower semicontinuity of the action, existence of geodesics, identification of the entropy slope, and convergence of minimizing movements.

12 Final answer

For the reversible jump flow of $f(\alpha) = \text{KL}(\alpha \mid \beta)$, the correct entropy-production condition is

$$\boxed{\frac{1}{2} \iint (r(y) - r(x))(\log r(y) - \log r(x)) \beta(\mathrm{d}x) K(x, \mathrm{d}y) \geq 2\kappa \int r \log r \, \mathrm{d}\beta, \quad r = \frac{\mathrm{d}\alpha}{\mathrm{d}\beta}.}$$

It is exactly the modified log-Sobolev inequality of the reversible jump generator and yields entropy decay $e^{-2\kappa t}$. It is also exactly the nonlocal Wasserstein–PL inequality when the logarithmic-mean action (9) induces a metric satisfying the slope identity (15); this extra metric identification should not be suppressed for singular target-dependent kernels.

This inequality is not implied by log-concavity, or even Gaussianity, of β without assumptions on K . It *is* guaranteed for a clean fractional model when one starts from a target satisfying an ordinary LSI and subordinates its reversible Langevin semigroup. For $\beta = \mathcal{N}(0, I_d)$ and generator $-(-L_{\text{OU}})^{s/2}$, one obtains the explicit Gaussian-reversible jump kernel (61) and the quantitative bound

$$\boxed{\kappa_s \geq 2^{s/2-1}.}$$

The pure translation-invariant fractional Laplacian, by contrast, has no Gaussian reversible probability target and should not be used to formulate this particular PL question without first modifying the generator.

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